

GALL BLADDER & BILIARY TREE — Clinical Revision Table

Bold = high-yield. Dr Mezjda Rashaan.

ANATOMY & INVESTIGATIONS		
Item	Key points	
GB	7–12cm, 30–50mL; Hartmann's pouch ; no submucosa/muscularis mucosa ; crypts of Luschka	
Cystic duct / artery	Valves of Heister ; cystic artery 90% from RHA in Calot's triangle ; Moynihan's hump danger	
CBD / Oddi	Opens at ampulla of Vater via sphincter of Oddi (relaxes with CCK)	
Investigations	US = standard (stones/jaundice); MRCP = gold standard biliary tree/CBD ; ERCP = therapeutic ; HIDA (acute cholecystitis/leak); CT for malignancy	
GALLSTONES		
Aspect	Detail	
Risk	5 F's (Fat, Female, Forty, Fertile, Family) ; F:M 3:1; haemolysis, ileal disease	
Types	Cholesterol (west, 80%, large/solitary, radiolucent); Pigment black (haemolysis/cirrhosis, in GB); pigment brown (infection/parasites, in ducts, Asia)	
Effects	GB: cholecystitis, mucocele, empyema, gangrene, carcinoma. Ducts: jaundice, cholangitis, pancreatitis. Bowel: gallstone ileus	
CHOLECYSTITIS & CHOLANGITIS		
Condition	Features	Treatment
Acute cholecystitis	Gallstone-blocked cystic duct; persistent RUQ pain, fever, Murphy's +ve , ↑WBC; US 95%	Conservative (IVF/antibiotics) → 90% respond → interval cholecystectomy ; non-response/gangrene → surgery
Chronic cholecystitis (biliary colic)	Pain 1–5h after fatty meal; normal WBC/LFT; US stones/sludge	Avoid fatty meals; cholecystectomy if diabetic/recurrent. Mucocele → palpable non-tender GB
Empyema / acalculous	Empyema = pus; acalculous = ICU/trauma/burns, 20% mortality	Drainage / cholecystostomy → cholecystectomy
Cholangitis	Obstruction + infection. Charcot's triad = pain+fever+jaundice ; Reynolds pentad +shock+confusion	IVF + antibiotics; 15% → biliary decompression (ERCP/PTC)
Choledocholithiasis	6–12% of gallstone pts; obstructive jaundice/pancreatitis. Courvoisier: palpable GB+jaundice ≠ stone (cancer)	MRCP dx ; ERCP + papillotomy
OTHER BILIARY DISEASE		
Condition	Key points	
Sclerosing cholangitis	Intra/extrahepatic stricture; 2/3 with UC ; cholangiocarcinoma 10–20%; ERC "beading"; stent/transplant	
Recurrent pyogenic cholangitis	Orient; bacteria + parasites → brown stones, strictures, abscess	
Parasites	Ascaris (via ampulla), Clonorchis (BD cancer) , Fasciola (water cress, eosinophilia, triclabendazole), hydatid	
Choledochal cyst	Female 6:1, premalignant → excision + Roux-en-Y	
Biliary atresia	1/4000; jaundice from birth; Kasai procedure ; untreated dies <3yr	
BD injury/stricture	Mostly iatrogenic (cholecystectomy) ; → Roux-en-Y hepaticojejunostomy	
BILIARY TUMOURS		
Tumour	Key points	
Cholangiocarcinoma	M=F, 50–70yr; risk PSC, UC, choledochal cyst, Clonorchis ; hilar = Klatskin (Bismuth-Corlette); painless jaundice + pruritus ; CA19-9; resectable 10–20%, median survival 18mo	
Gallbladder carcinoma	F:M 5:1 ; >90% gallstones ; adenocarcinoma; risk stones >3cm (10×), polyp >10mm, porcelain GB ; spreads to liver early, 40% metastatic at dx; cholecystectomy ± hepatectomy by depth	